

Name: _____

Python Tutorial Assessment

BMI 500: Introduction to Ethical Data Science and Informatics

Please complete the tutorial <https://www.learnpython.org/>. Please complete this assessment in a Jupyter notebook or on Google Colab.

The Leibniz formula¹

$$\frac{\pi}{4} = \arctan 1 = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$$

is one of the first formulas for calculating π that was based on infinite series instead of geometrical techniques. While this series converges much more slowly than many other series, it can still provide several digits of π within seconds.

Problem 1. Write a function in Python 3.x that, given an integer n , returns the sum of the first n terms of the series in the Leibniz formula.

Problem 2. For each of the following items, write a function in Python 3.x that, given an integer n , returns the sum of the first n terms of the series in the Leibniz formula.

- Use a for-loop and an if-statement with the modulo operator `%` to determine whether to add or subtract each term.
- Use a for-loop with the quantity `(-1)**n` to determine whether to add or subtract each term.
- Construct a Python list and compute the sum of the terms in the list.
- Construct a Python set and compute the sum of the terms in the set.
- Construct a Python dictionary and compute the sum of the terms in the dictionary.
- Construct a NumPy array and compute the sum of the terms in the array.
- Construct a NumPy array, use array indexing to compute the sum of the positive terms in the array, use array indexing to compute the sum of the negative terms in the array, and add the two sums together. You can write `x[: :2]` to access the first, third, etc. terms and `x[1: :2]` to access the second, fourth, etc. terms.
- Combine the first and second terms, the third and fourth terms, etc. to change this series from an alternating to a non-alternating series and compute the sum of the combined terms.

¹The Leibniz formula is a special case of more general formulas that we first discovered by Madhava and Gregory.

Problem 3. Which of these implementations of the Leibniz formula is the most accurate, fastest, and/or clearest? Which function would you use to calculate π (and why)? You can use the built-in constants `math.pi` or `np.pi` for assessing the accuracy of your functions and the `time` package or the `%timeit` command for timing.

Problem 4. Choose your favorite implementation of this formula, and plot the absolute error in the sum as a function of the number of terms in the sum.

You should have accurate, informative, and legible labels for both the x -axis and y -axis, and you should use logarithmic axes or take the logarithm of the errors and the numbers of terms so that you can see the sums converge over a wide range of values.

By default, most plotting packages generate large plots that are difficult to read in papers and presentations. Use a command like `plt.figure(figsize=(2, 2))` to make your plot more legible.

Problem 5. Would you do anything differently for computing π using the Leibniz formula if you were using MATLAB instead of Python?